Dynamic models of systems and standard forms for system models

R0005E - Measurement and Feedback Control

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First principle models

Background

First principle models are based on the laws of e.g. physics of a process and describe how a process behave both from a dynamic and a static perspective.

Usually, this can be time consuming and very often it is necessary to decompose your process into interconnected elements.

Decomposition

- Identify basic elements: $\Rightarrow$ Element laws
- Determine how the elements interact: $\Rightarrow$ Interconnection laws
- Identify Variables

Combining the resulting equations will result in a first principle model and is a formulation in mathematical terms. These equations can be classified according to several criteria:

- Spatial characteristics: lumped or distributed.
- Continuity of the time variable: Continuous, discrete or hybrid.
- Quantization of the dependent variable: Quantized or non-quantized.
- Parameter variation: Fixed or time-varying.
- Superposition property: linear or non-linear.

Question:

How much can we trust a model? Does a model exactly represent real-life? Well, there is a well-known quote that gives the answer:

> 'Essentially, all models are wrong, but some are useful.'
>
> by George E.P. Box

Some reasons:

- Some facts are simply not known or understood.
- When models are derived, assumptions are made.
- Some facts need to be approximated.
- ...

Translational mechanics

Typical Variables:

- x: Displacement, SI-Unit: m
- v: Velocity, SI-Unit: m/s
- a: Acceleration, SI-Unit: m/s^2
- f: Force, SI-Unit: N

Element laws:

- Mass: $f = \frac{d}{dt}(Mv)$, where M denotes the mass. If the mass is constant then this renders into $f = Ma$.
- Friction: $f = B\Delta v$, where B is the friction coefficient.
- Stiffness: $f = K\Delta x$, where K is the coefficient that converts displacement into a force.

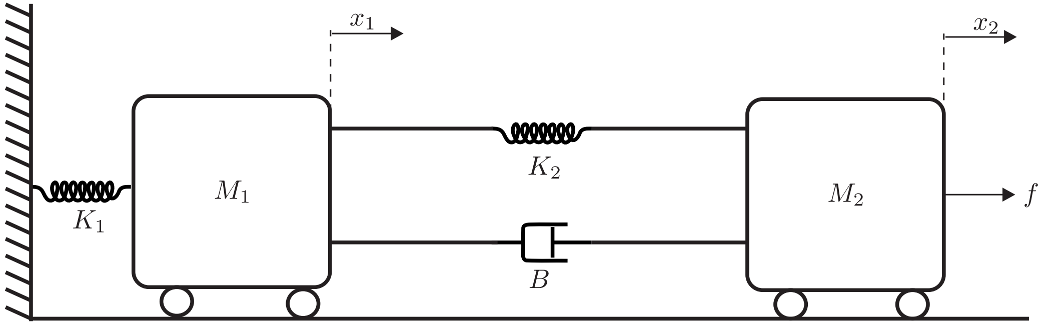
Interconnection laws:

- D'Alembert's law: $\Sigma_i f_i = 0$
- Law for displacement: Connection points move with the same speed and have the same displacement.
- Law of reaction forces: Every force of one element on another is accompanied by a reaction force in the opposite direction.

Translational mechanics (2)

Task:

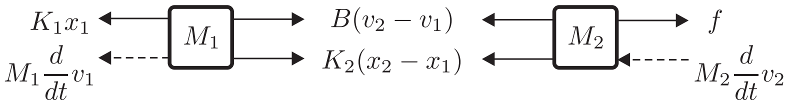
Derive a first principle model of the system given in the following schematic:



This two mass system can represent many different technical systems and could be a car suspension, computer hard disk drive..

Decomposition:

First we decompose the system using the element laws and derive a free body diagram:



There, $v_1 = \frac{d}{dt} x_1$ and $v_2 = \frac{d}{dt} x_2$.

Note: We will from now on denote these time derivatives as follows: $v_1 = \dot{x}_1$ and $v_2 = \dot{x}_2$.

Composition:

Now we can apply D'Alembert's law to the equations in the free body diagram:

$$\begin{aligned} -M_1\dot{v}_1 - K_1x_1 + K_2(x_2 - x_1) + B(v_2 - v_1) = 0 \\ -M_2\dot{v}_2 - K_2(x_2 - x_1) - B(v_2 - v_1) + f = 0 \end{aligned}$$

Clearly, the model is composed of 2 differential equations of order 2. The complete model is therefore of order 4.

Some observations:

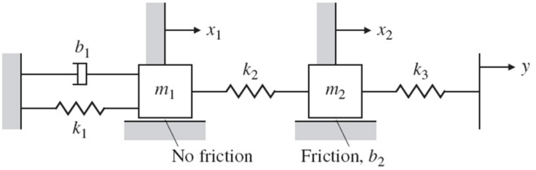
- The inputs to the system are defined in the system design. Here it is f.
- Outputs can be chosen freely. The outputs are points of observation. E.g. acceleration in M_2 or position of M_1.
- If $f = 0$ and the input would be x_2, then $\dot{x}_2$ and $\ddot{x}_2$ are known.
  Thus, the model order can be reduced.
- If the parameters $(K_1,\ K_2,\ ...)$ are known, then the system is completely defined.
- If some parameters are unknown, then experiments can be conducted to determine the parameters.

Recommended problem

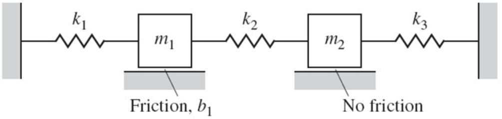
Now we can practice first principle modeling. Perform the following steps on the problem 2.1 in the textbook:

- Identify the elements and the interconnections
- Identify the typical variables that are present in the systems
- Create the free body diagrams
- State the differential equations or set of equations

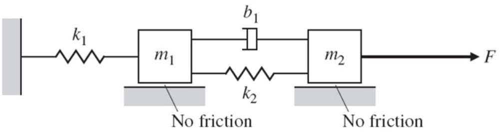
Sketch for the systems:



(a)



(b)



(c)

Standard forms: Input-Output form

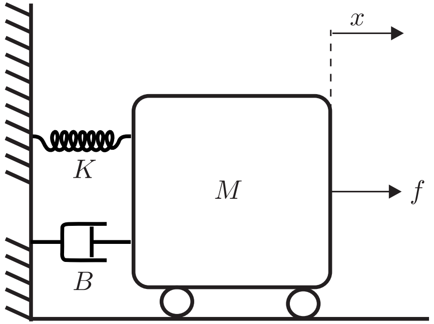
The input-output form is a reformulation of the differential equation of a system and is given as

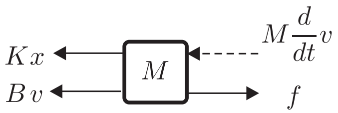
$$a_n y^{(n)} + \cdots + a_2 \ddot{y} + a_1 \dot{y} + a_0 y = b_m u^{(m)} + \cdots + b_1 \dot{u} + b_0 u$$

Thus, find the output variable and its derivatives and the input and its derivatives. The output variable is always an unknown and the equations have to be solved for that equation.

Example:

Find the input-output form of the system below with displacement x being the output and the force f being the input.





Solution:

First we derive the first principle model from the free body diagram and the application of D'Alembert's law:

$$-M\ddot{x} - B\dot{x} - Kx + f = 0$$

This can be reformulated into

$$M\ddot{x} + B\dot{x} + Kx = f$$

Now we can introduce $y \triangleq x$ and $u \triangleq f$ and get the input-output form:

$$M\ddot{y} + B\dot{y} + Ky = u$$

But: In the first example with two connected masses it is not that easy to derive the input-output form, since the 2 differential equations are connected. Remember:

$$\begin{aligned} -M_1\dot{v}_1 - K_1x_1 + K_2(x_2 - x_1) + B(v_2 - v_1) &= 0 \qquad (1) \\ -M_2\dot{v}_2 - K_2(x_2 - x_1) - B(v_2 - v_1) + f &= 0 \qquad (2) \end{aligned}$$

Assume, output $y \triangleq x_2$ and input $u \triangleq f$.

Trick: Introduce the differential operator s as follows:

$$s \triangleq \frac{d}{dt} \quad \Rightarrow \quad \begin{aligned} sy &= \dot{y} \\ s^2y &= \ddot{y} \\ &\vdots \end{aligned}$$

Now, the input-output form can be rewritten into:

$$a_ns^ny + \cdots + a_2s^2y + a_1sy + a_0y = b_ms^mu + \cdots + b_1su + b_0u$$

This is now just an algebraic equation in y and u, and can be solved for y:

$$y = \frac{b_ms^m + \cdots + b_1s + b_0}{a_ns^n + \cdots + a_2s^2 + a_1s + a_0}u$$

The term

$$\frac{b_m s^m + \cdots + b_1 s + b_0}{a_n s^n + \cdots + a_2 s^2 + a_1 s + a_0}$$

is called a *transfer function*. A more thorough derivation of a transfer function and the operator s will be given later in the course. For the time being, we can say that the transfer function describes in what way an input generates an output for a system.

Note :

- s is not a variable, it is an operator and should not be solved for.
- The numerator and denominator are polynomials in s.
- The transfer function captures the behavior of the system.
- In Matlab the command `tf` can be used to represent a transfer function.

Task: Derive the input output form and transfer function for the system described by (1) and (2).

Standard forms: State variable form

Instead of resolving all the model into one equation in the input output form, one could think of representing the model as a set of equations, where each equation has the order one.

This form is called the state variable form.

Remember the mass-spring-damper system from before:

$$M\ddot{x} + B\dot{x} + Kx = f$$

This is one second order equation. Now we could introduce one additional equation:

$$\dot{x} = \frac{d}{dt}x$$

In the next step we introduce state variables:

$$\begin{array}{rcl} q_1 & = & x \\ q_2 & = & \dot{x} \end{array} \quad \Rightarrow \quad \begin{array}{rcl} \dot{q}_1 & = & \dot{x} \\ \dot{q}_2 & = & \ddot{x} \end{array}$$

Now we can replace the variables in the model with the state variables and get two equation:

$$\begin{aligned} \dot{q}_1 &= q_2 \\ \dot{q}_2 &= -\frac{K}{M}q_1 - \frac{B}{M}q_2 + \frac{1}{M}f \end{aligned}$$

Standard forms: State variable form (2)

Some observation:

- On the left hand side there are only first order derivatives and on the right hand side there are only state variables
- This is a set of first order differential equations
- It can easily been solved and is useful for computer simulations

Note: The state variable form will not be used in this course but in the continuation course R7003E. More information can be found in chapter 7.

Rotational mechanical systems

Typical Variables:

- θ: Angular displacement, SI-Unit: rad
- ω: Angular velocity, SI-Unit: rad/s
- α: Angular acceleration, SI-Unit: rad/s^2
- M: Torque, SI-Unit: Nm

Element laws:

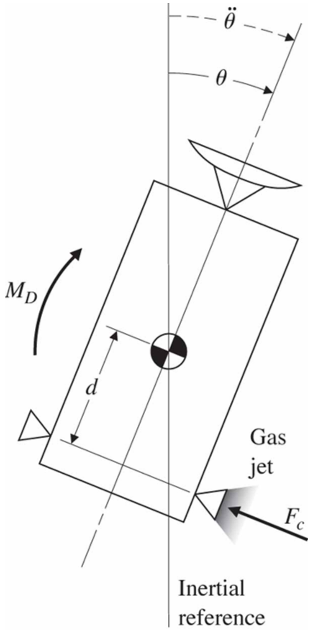
- Moment of inertia: $M = \frac{d}{dt}(I\omega)$, where I denotes the moment of inertia. If the moment of inertia is constant then this renders into $M = I\alpha$.
- Friction: $M = B\Delta\omega$, where B is the rotational friction coefficient.
- Stiffness: $M = K\Delta\theta$, where K is the coefficient that converts displacement into a torque, e.g. torsional spring.

Interconnection laws:

- D'Alembert's law: $\Sigma_i M_i = 0$
- Law for angular displacement: Connection points move with the same angular speed and have the same displacement.
- Law of reaction torques: Every torque of one element on another is accompanied by a reaction torque in the opposite direction.

Task:

Derive a first principle model for the motion of a satellite in space given by the following schematic:



Source: Feedback Control of Dynamic Systems, Franklin et. al.

Solution:

First the system should be decomposed using the element laws and interconnection laws.

In this case the free body diagram is more or less the same as the picture to the left. It can be observed that there is an additional torque M_D, which is a disturbance acting on the satellite.

Rotational mechanical systems (3)

Thus, there are two inputs to the system, namely M_D and F_C. The latter is the forced that is used to control the satellite.

Applying D'Alembert's law yields the following equation:

$$F_C d + M_D - I\ddot{\theta} = 0$$

Some observation:

- Same procedure as for translational mechanical systems
- Obviously, a combination of both systems will be possible
- Standard forms can be applied in a similar way.

Input-Output form and s-operator

In the textbook by Franklin, the term transfer function is introduced using the solution of a first order differential equation. This will actually be presented more precise in chapter 3.

Alternative derivation:

Remember the s-operator from before, which represents the derivative of a variable. Then, the inverse of s, namely $\frac{1}{s}$ represents the integration of a variable.

In case of the satellite motion, the differential equation is given as

$$F_C d + M_D - I\ddot{\theta} = 0$$

and can be reformulated to

$$\ddot{\theta} = \frac{F_C d + M_D}{I} \tag{3}$$

Applying the s operator yields

$$\theta = \frac{1}{s^2} \frac{F_C d + M_D}{I} \tag{4}$$

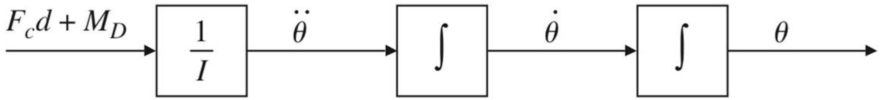
Conclusion: The angular displacement is given by the double integration of the angular acceleration.

Note:

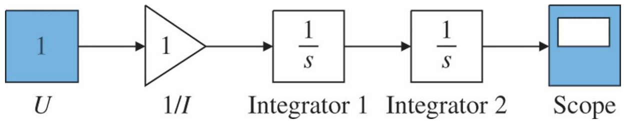
- In the textbook s is introduced which is also known as the Laplace operator.
- Initial conditions are not considered here!

Blockdiagrams

The differential equation 3 can be represented in a block diagram:



Clearly, the output of the block diagram is the solution to the differential equation. Thus, a numerical solution can be found in this way. An implementation of the motion in SIMULINK, which is a simulation environment within Matlab, is given as follows:



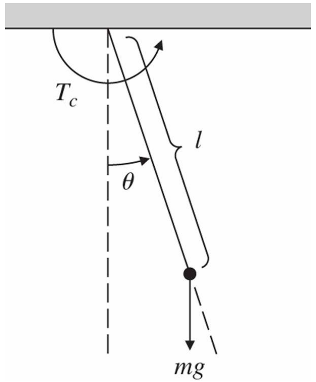
Try, to choose different input signals like a Step function or a square pulse function. Blocks that generate these kind of signals are found in the library called *Sources*. For example, the use of the square pulse function would mean that the gas jet would be fired for a certain amount of time.

Important notes:

- You need to choose a value for I
- This is not a closed-form solution to the problem.
- In this form, initial condition can not be considered. This will be discussed later!
- Simulink has the ability to solve the problem with initial conditions. Those have to be set in the integrator block.

Example: The pendulum

A typical example of a rather simple non-linear mechanical system is the pendulum:



The parameters of the pendulum are mass $m = 1kg$, the length $l = 1m$ and $g = 9.81m/s^2$.

Model

First we derive the equations of motion. Here we could use two approaches:

- Translational mechanics:
$$F_c - mg\sin\theta = ml\ddot{\theta}$$

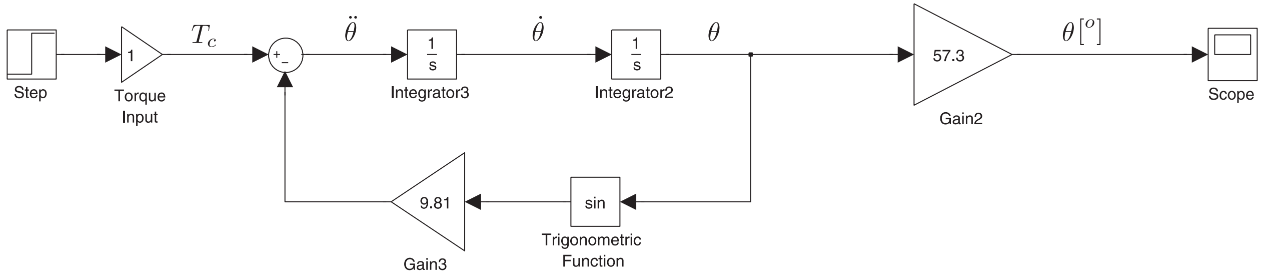
- Rotational mechanics:
$$T_c - mgl\sin\theta = I\ddot{\theta}$$

Both equations can be rewritten into the form

$$\ddot{\theta} + \frac{g}{l}\sin\theta = \frac{T_c}{ml^2} = \frac{F_c}{ml} \tag{5}$$

Block diagram

As before, the equation (5) can be expressed as a block diagram. We do this directly in SIMULINK.

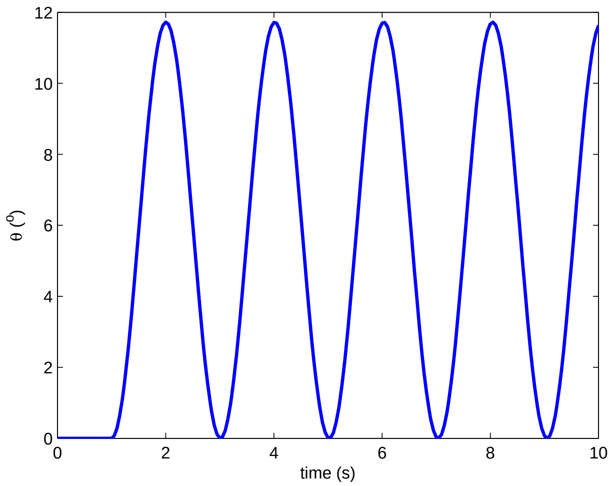


Now we can assume that a constant torque $T_c = 1Nm$ will be applied from time $t = 1s$ and forward. This can be expressed by a step function:

$$T_c = \begin{cases} 0Nm, & t < 1s \\ 1Nm, & t \geq 1s \end{cases}$$

Simulation

Now we can simulate the behavior of the pendulum and observe how the system responds to the step function and how the angle of the pendulum behaves over time. In other words it is the numerical solution to the differential equation.

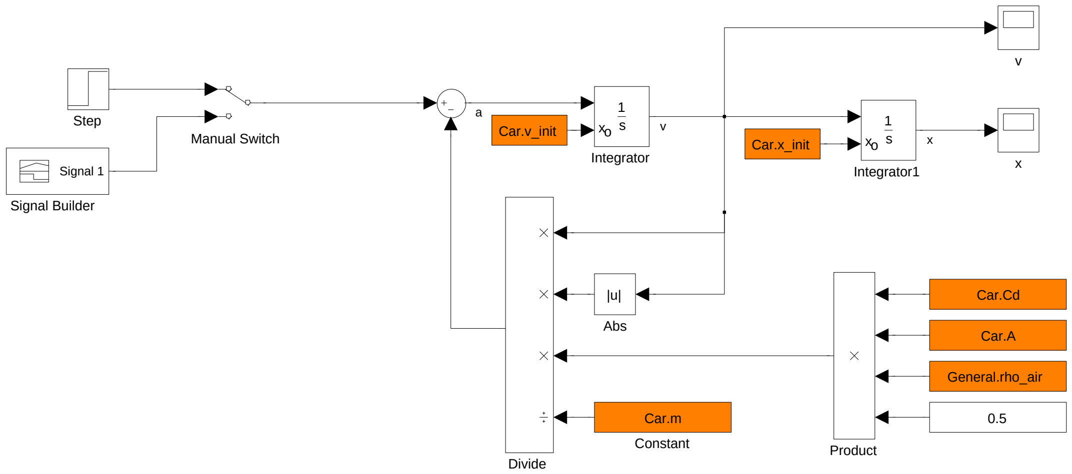


Some Observations:

- The pendulum exhibits an undamped oscillation.
- The pendulum oscillates around angle angle close to 6^o.

Example: Realistic car

Assume a car is moving along a straight road with a road grade of 0^o. But we will consider the aerodynamics of the car. A simulation model can be derived according to the following block diagram:



In order for the model to work we need the parameters that are given in the block with the orange background. These are both initial conditions and parameters for the model.

Note:

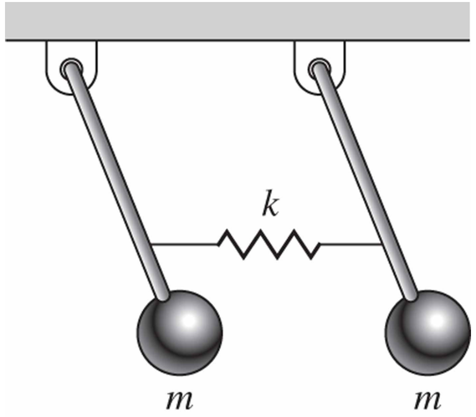
- The model is provided to you in CANVAS. You can experiment with it! Analyze the behavior of the car for the different inputs.
- You need to load the parameter file $car\_ini.m$ prior to a simulation.

Recommended problem

Now we can practice first principle modeling for a system which exhibits properties of both translational and rotational mechanics. Perform the following steps on the problem 2.3 in the textbook:

- Identify the elements and the interconnections
- Identify the typical variables that are present in the systems
- Create the free body diagrams
- Write down the differential equations and combine them into two equations.

System sketch:



Further problems

Later in the course and in the lab assignments, we will make use of the software Matlab/Simulink. The problems 2.5 and 2.8 make use of Matlab.

If you are new to Matlab, then there are two sources:

- There is a very good tutorial on the internet for Matlab/Simulink: `http://ctms.engin.umich.edu/CTMS/index.php?aux=Home`
- Use the Getting started and help/tutorial features in Matlab.

Hints:

- There are some Matlab code snippets in the textbook. Re-use them.
- Represent the systems as a transfer function in Matlab using the command `tf`
- Make use of the Matlab function `step`. This function calculates the output of the system as a function of time when the input changes from one value to another momentarily (step-like change).
- Analyse the plots. Remember that the plot describes the output for a step of height 1 from an initial value 0.

Electric circuits

Typical Variables:

- v: Voltage, SI-Unit: V
- i: Current, SI-Unit: A

Element laws:

- Resistor: $v = Ri$.
- Capacitor: $i = C\frac{dv}{dt}$.
- Inductor: $v = L\frac{di}{dt}$.
- Voltage source: $v = v_s$.
- Current source: $i = i_s$.

Interconnection laws:

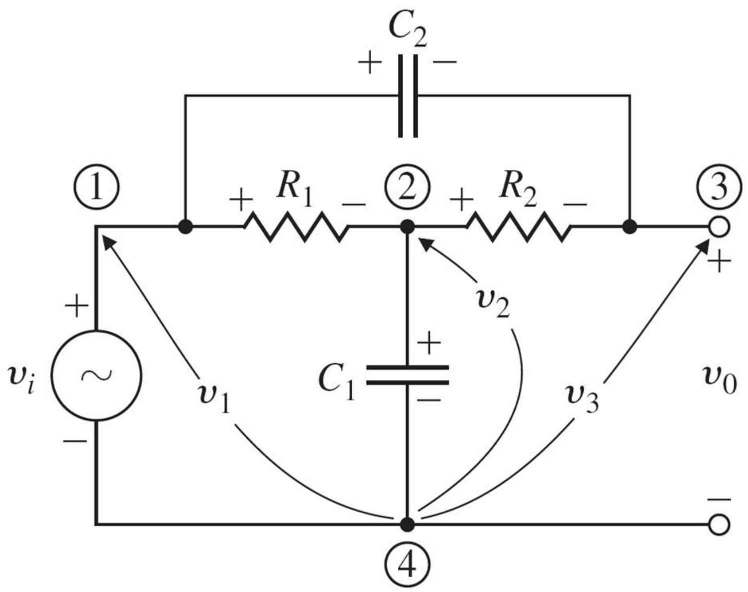
- Kirchhoff's current law: Sum of all currents leaving a junction is equal to the sum of the currents entering a junction.
- Kirchhoff's voltage law: Sum of all the voltages along a closed path is equal to zero.

Node analysis

In order to derive a model for an electric circuit a node analysis is performed and all the equations are derived for the nodes:

1. Determine the number of nodes N
2. Select one node as the reference node, preferably the ground or common level for the circuit.
3. For the remaining $N-1$ nodes derive the node equations.

Bridge Tee Circuit:



Node analysis (2)

Now we have selected the nodes and denoted the voltages at the nodes. Obviously, the following holds:

$$v_1 = v_i \qquad v_o = v_3$$

Here, v_i is the known input and thus, v_1. Therefore we only need another two equation to fully describe the system. For node 2 we get:

$$-\frac{v_1 - v_2}{R_1} + \frac{v_2 - v_3}{R_2} + C_1 \frac{dv_2}{dt} = 0$$

Finally, we assess node 3:

$$\frac{v_3 - v_2}{R_2} + C_2 \frac{d(v_3 - v_1)}{dt} = 0$$

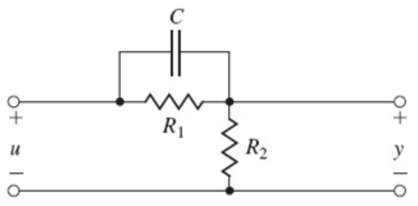
Note:

- We could now resolve for v_o for a given input voltage v_i. This is then the input-output form.
- In this case, one of the node equations disappeared. This is not always the case! It was due to the fact that the voltage source was directly connected between the reference node and another node.
- Using the s-operator in the element laws directly, will render algebraic equations instead. This is also the preferred way when deriving models for electrical circuits.

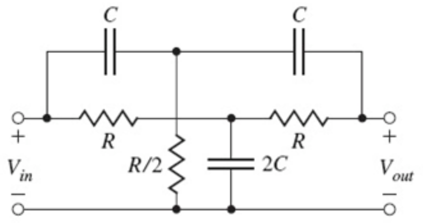
Recommended problem:

Make use of the node analysis in the problems 2.15a and 2.15d. You can also implement the models in SIMULINK and experiment with the different input signals.

Sketch of the circuits:



(a)



(d)

Electromechanical systems

Electromechanical systems are of large importance. In that case mechanical systems are connected to electric circuits and there is an element that creates the interface. These are mostly motors and generators.

Additional element laws:

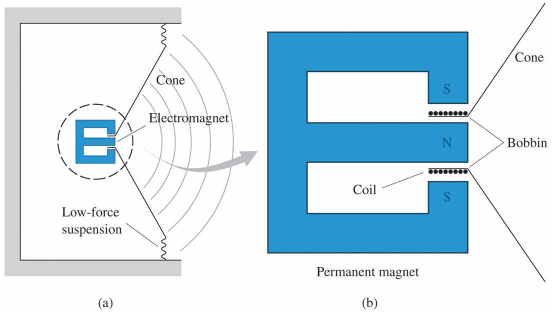
- Law of motors: $F = Bli$, where B is the magnetic field strength, l is the length of the conductor and i the current in the conductor.
- Law of generators: $e = Blv$, where B is the magnetic field strength, l is the length of the conductor and v is the speed of the conductor in the magnetic field.

Note:

Here, e denotes the voltage in $[V]$ and v denotes the speed in $[m/s]$. Be careful with the notation when you derive dynamic models for electromechanical systems.

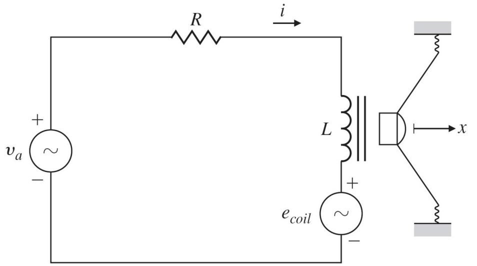
Example: loud speaker

The principle of a loud speaker can be given by the following sketch.



The motion of the cone will create acoustic wave which we can perceive as sound. The motion of the cone directly depends on the input current and of the constructive parameters of the speaker hardware.

The loud speaker can be represented by the following electric circuit. Clearly, the generation of the wave is not captured here.

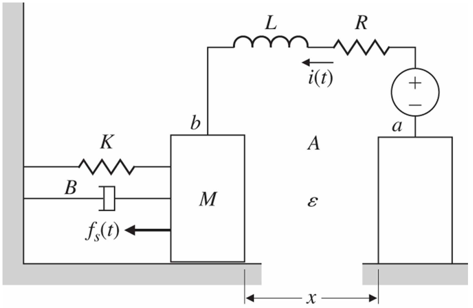


The motion of the cone will also induce a voltage e_{coil} into the circuit. Thus, there is a bi-directional effect.

Recommended problem:

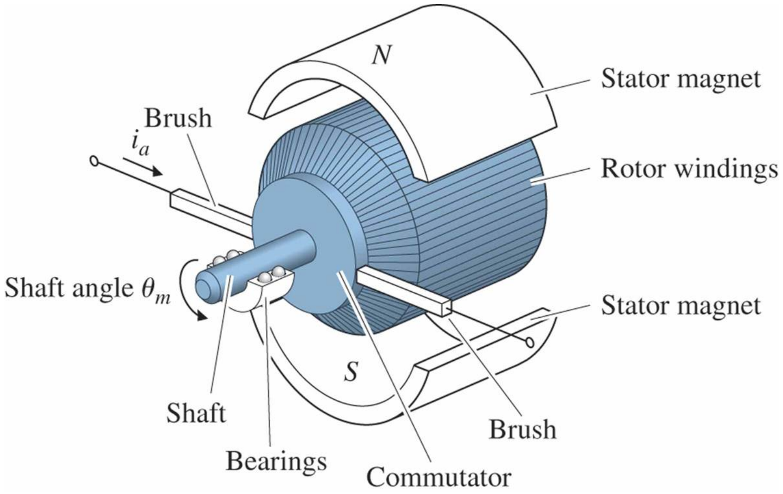
Derive the model for the capacitor microphone in problem 2.19a and determine output as requested in 2.19b.

Sketch of the system:



Example: DC motor

The DC motor can be used as an actuator in many applications. It is therefore of large interest. Just think of Hybrid car, which most car manufacturer offer their customers, there are motors and generators in the car besides the normal combustion engine.



The motor current will create a torque on the shaft and the shaft speed will create a voltage in the electric circuit:

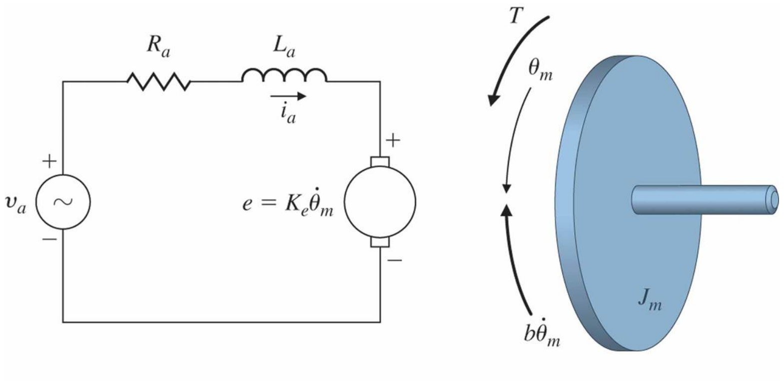
$$T = K_t i_a \tag{6}$$

$$e = K_e \dot{\theta}_m \tag{7}$$

There, K_t is the torque constant of the motor and K_e is the electrical constant of the motor.

Example: DC motor (2)

For the electrical part a circuit diagram (a) can be created and for the mechanical part a free body diagram (b) can be produced:



(a) (b)

Now we can derive an dynamic model for the mechanical part:

$$J_m\ddot{\theta} + b\dot{\theta} = K_t i_a \quad (8)$$

where J_m is the inertia of the rotor and b is the friction coefficient of the rotor.

Similarly, a dynamic model of the electric circuit can be derived:

$$L_a\frac{di_a}{dt} + R_a i_a = v_a - K_e\dot{\theta}_m \quad (9)$$

where L_a is the inductance of the motor and R_a is the resistance of the motor.

Equations (8) and (9), represent a complete dynamic model of the motor.

Recommended problem:

Derive a model for a motor in problem 2.20. Additional question: What are inputs and outputs?

Sketch for the system: